

Monopile Installation Record HB-A02

Harrow Bay Offshore Wind · Foundations

Elena Márquez · 14 April 2026 · HB-CON-IR-A02

Position	Date	Vessel	Contract
HB-A02	12 Apr 2026	MV Sea Atlas	WE-CON-2025-040 / -027

1. Position and pile data

Item	Design	As-installed
Coordinates (WGS84)	53.91410 N, 1.04780 E	within 36 cm of target
Water depth (m LAT)	29.3	29.3
Pile length (m)	72.8	72.8
Pile mass (t)	1,253	–
Target penetration (m)	34.5	34.6
Verticality	≤ 0.25°	0.09°

2. Driving record

Item	Value
Start of soft start	13:50
Driving duration	4 h 08 min
Total blow count	5,120
Maximum hammer energy	2,551 kJ
Self-weight penetration	5.3 m
Interruptions	1 × 40 min (bubble curtain compressor trip)

3. Noise and marine mammal mitigation

Item	Value
Bubble curtain	Double big bubble curtain, both rings operational
Peak SEL at 750 m	157.9 dB (limit 160 dB)
Pre-piling search (MMO + PAM)	30 min, marine mammal detected – start delayed 30 min
Acoustic deterrent device	Activated 30 min before soft start

4. Transition piece

Item	Value
------	-------

TP installed	14 Apr 2026
Flange flatness after installation	1.3 mm (limit 2.0 mm)
Bolt tensioning	148 × M72 – completed
TP level	+0.01° (limit 0.25°)

5. Non-conformances

No non-conformances raised.

6. Sign-off

Role	Name	Date
Offshore Installation Supervisor	Atlas Marine Contractors	13 Apr 2026
Client Representative	Felix Brandt	13 Apr 2026
Reviewed	Elena Márquez	14 Apr 2026